# CARGO MATE
# SMART ROBOTIC INVENTORY SYSTEM WITH CLOUD INTEGRATION

**25-26J-032**

Project Proposal Report


B. Kethmi Senara

B.Sc. (Hons) Degree in Information Technology specialized in Computer Systems and Network Engineering

Department of Computer Systems Engineering

Sri Lanka Institute of Information Technology
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# Declaration of the Candidate & Supervisor

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any previously published or written by another person material, except where the acknowledgement is made in the text.

| Group Member Name | Student ID | Signature |
|---|---|---|
| Senara B K | IT22036698 | |

The above candidate are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the Supervisor :

Date 30/08/25

# Abstract

This report focuses on the development, design, and integration of the Robotic Arm Process in a Smart Robotic Warehouse Inventory System. The robotic arm is a focal point of the system for automated object handling, like stock-in (warehousing products) and stock-out (withdrawing products) activities. It is complemented by a mobile trolley that transports goods within the warehouse and a cloud-based inventory management system that contains real-time histories of stock status, location, and movement.

Inclusion of the arm in warehouse operations allows repeat and precise handling of the products, with fewer chances of human error that comes with manual processes. The arm is equipped with grippers and sensors to pick and place goods in precise identification, even in dense storage environments. The arm is also optimized to carry out optimal pick-and-place routes that result in reduced movement time and maximum throughput overall.

The warehouse has real-time tracking and monitoring of all the products in stock as the robotic arm is connected to the cloud-based inventory system. Automatic updating of stock databases when storing or collecting products, allowing accurate reporting, efficient space allocation, and quick decision-making in inventory management are facilitated through this connection. Predictive reporting and analytics are also offered by the system, allowing warehouse managers to predict demand, plan for replenishment, and reduce operating constraints.

Overall, the Robotic Arm Process augmented by mobile trolleys and cloud inventory management represents a comprehensive automation system. It enhances working efficiency, offers accurate and timely goods handling, and is flexible to the needs of advanced logistics through an extensible and intelligent warehouse system.

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# 1. Introduction

Today's warehouses are increasingly turning to automation to maximize efficiency, precision, and scalability of the inventory management process. As higher volumes of products and customer requirements for faster delivery, traditional manual procedures are not effective. Manual handling is cumbersome, human-error prone, and non-repeatable, leading to imbalance in stocks, misplaced items, and late delivery of orders. Such issues are being addressed by warehouses by adopting intelligent inventory systems with autonomous robots, cloud-based computing platforms, and IoT sensors for automating the logistics process, improving accuracy, and optimizing warehouse performance.

This project's Smart Robotic Warehouse Inventory System is a great example of this paradigm shift. It has been developed as a holistic automation framework comprising multiple smart elements that collaborate to manage inventory efficiently. Upon which lies, this system is a trolley with IR sensors, RFID readers, and ultrasonic detectors. Sensors enable the trolley to travel autonomously along warehouse corridors, detect intersections, negotiate around obstacles, and transport loads to and from the robotic arm.

Lying at the center of the system is a cloud-based inventory system that is achieved through Firebase. The platform maintains real-time inventory levels, tracks locations of items, sends automatic restocking alerts, and provides precise storage or retrieval orders to the robotic arm and cart. The cloud system is the primary control center that directs the movements of the robots as well as updates inventory data in real-time.

Barcode reading integration allows the system to read individual products, tag their status as updated, and keep the inventory database current at all times. Additionally, space location detection is accomplished through weight sensors, which detect space locations available and maximize space utilization.

The focal point of this setup is the robotic arm module, which automates the stock-in (storing) and stock-out (retrieval) functions. During stock-in procedures, the arm removes items from the mobile trolley and loads them into predestinated rack positions with precision. During stock-out procedures, the arm removes items from the racks and transfers them onto the trolley for shipping. The cloud system controls the arm, makes exact pick-and-place motions, and updates the real-time inventory database.

The automation of the robotic arm with cloud-based platform and trolley ensures seamless coordination of activities across the warehouse. Automated item handling reduces human intervention, errors, and optimizes operating speed. By utilizing real-time monitoring, intelligent navigation, and precise robotic manipulation, this Smart Robotic Warehouse Inventory System is an effective and scalable system for warehouse management today, overcoming problems of large volume and rapid order processing.

## 1.1. Background and Literature Survey

### 1.1.1. Background

Industrial automation has been based on robotic arms that have been largely used in manufacturing, assembly lines, and packaging. The principal advantage of robotic arms is that they can perform repetitive, accurate, and boring tasks more quickly and evenly compared to human labor. Robotic arms cut down variability and fatigue to improve industrial quality and productivity.

Over the last few years, they have been used for applications outside traditional manufacturing in warehouse automation, where they are employed in the critical function of optimizing inventory handling and logistics operations.

A robotic arm in a warehouse environment is the primary manipulator in an automated system responsible for moving goods between heterogeneous components such as mobile trolleys, storage racks, and dispatch areas effectively. Its main function is:

Stock-In (Storage): The robotic arm transfers items from a delivery trolley to assigned storage racks, putting them in place precisely by location and category. This reduces the risk of lost items and maximizes storage capacity.

Stock-Out (Retrieval): The robotic arm retrieves the items automatically upon an order for retrieval, and they are placed on a trolley for shipping out. It decreases effort and time spent in order fulfillment.

Inventory Confirmation: Robotic arms can be combined with barcode readers, RFID readers, or vision-based systems to identify and locate every item precisely and put it in the right location, with precise real-time inventory tracking.

Through carrying out these vital tasks with automation, robotic arms cut down on human error, reduce physical strains on warehouse staff, and allow for real-time syncing with cloud-based stock management. This coordination enables accurate levels tracking of stocks, optimizes warehouse operations, and enables expandable high-volume logistics. Application of robotic arms in warehouses is a key innovation in intelligent warehouse technology that combines precision, efficiency, and smart coordination to meet existing inventory management challenges.

### 1.1.2. Literature Survey

In recent years, the deployment of robotic arms has greatly enhanced the automation of warehouse processes. Typical robotic arm operations include industrial pick-and-place operations, but there are many more applications to include inventory operations and warehouse navigation. Various studies have evaluated different components of arms, including gripper design, object detection, and placement accuracy.

Sobhan and Shaikat [1] evaluated arms specifically in warehouses. The authors highlighted that the arms demonstrate continued potential to automate repetitive tasks in warehouses. But they pointed out that low grip reliability, due to a lack of adaptive gripping surfaces, often resulted in objects slipping or being mishandled.

McTaggart et al. [2] have offered the advantage of improved placement accuracy by researching placement with Cartesian manipulators and sight-based vision systems. However, the authors relied solely on a three-dimensional viewing system that only allowed for occasional drop placement errors and often struggled when the handled products were reflective surfaces or were irregular shaped objects.

Similarly, Beul et al. [3] evaluated automated inventory updates with autonomous systems but couldn't provide real-time feedback once the objects had been handled and replaced. So, there is an increased risk of mishandling or dropping items in pick-and-place operations, when placing objects into inventory tracking systems.

Two-dimensional navigation and all various combinations of systematic navigation and coordination were covered by Han et al. [4] investigating a robotics arms ability to path plan efficiently. However, no attention was directed to the interaction between the gripper or the arm, and the various objects they handle, leading to reduced operational handling of fragile or warped goods.

Finally, Bhatia et al. [5] studied the design of gripper systems for accommodating different types of objects. They improved adaptability through design changes, but the limitations of the material and lack of force feedback on surfaces meant that the system couldn't safely and reliably handle many surfaces.

According to other recent studies, these limitations could be addressed by introducing force sensing, tactile feedback and accurate distance measurement. For example, FSRs (force sensitive resistors) can be used to ensure appropriate grip force is applied to avoid slippage or damage to the object being gripped or placed, while TOF (time of flight) sensors can accurately measure the distance between the gripper and the object being manipulated to ensure accurate placement. Additionally, applying a rubber pad at the distal end of the gripper can increase friction and stability and, thus, allow robotic arms to manipulate objects with various sizes and textures in a safe manner [1]– [5].

## 1.2. Research Gap

This research investigates enhancing warehouse robotic arm systems through grip reliability, adaptive force sensing, and precise placement. Although the previous work [1]–[5], Bhatia et al., and Han et al. evaluated pick-and-place automation, vision based placing, and gripper design, they did not take a holistic approach relying on tactile feedback, friction improvement, and distance measurement. The proposed system solves these issues by using rubber pads for improved grip reliability, force sensing rubber pads through FSR sensors to monitor force, and employing a TOF sensor to place items precisely to improve automation within warehouse operations while reducing handling errors.

*Table 1. 2. 1 Research Gap Table*

| Research Paper | Focus/Contribution | Limitation | Proposed Solution |
|---|---|---|---|
| Implementation of Pick & Place Robotic Arm for Warehouse Products Management | Warehouse robotic arm for pick-and-place | Low grip reliability, no force sensing | FSR is used with a rubber pad for grip reliability; FSR also provides force information |
| Adaptive Intelligence in Warehouse Robotics: Efficient Pick-and-Place Robot for Dynamic Environments | Adaptive intelligence in warehouse robotics | Heuristic process that uses camera only for object placement detection | Attached TOF sensor allows for precise distance measurement leading to accurate object placement |
| Design and Development of an Automated Robotic Pick & Stow System for an e- | Automated pick & stow system for e-commerce warehouses | Use autonomous systems, but no feedback regarding object handling | FSR provides continuous feedback on force; TOF feedback provides |

| | | | |
|---|---|---|---|
| Commerce Warehouse | | | confidence on object placement |
| Automation of a Warehouse by Means of a Robotic Arm | Warehouse automation with robotic arm | No consideration of gripper and object interaction with gripper attachments | Rubber pad improves gripping ability; FSR provides safe gripping force feedback |
| Adaptive Intelligence in Warehouse | Adaptive intelligence in warehouse robotics | Gripper material used leads to poor adaptability; did not utilize continuous feedback to force intervention during handling | Using a rubber pad increases the maximum friction force available for interacting with various surfaces; placement also adapted using TOF sensor used to inspect and react during placement process. |

## 1.3. Research Problem

While there have been many advances in warehouse automation, robotic arm systems still have critical issues in being able to reliably deal with variability in objects and provide reliable place locations. In the literature, Sobhan and Shaikat [1], Ganduri et al. [2], and numerous others have demonstrated pick-and-place automation with adaptive intelligence in warehouse environments. However, most of these systems have relied on feedback methods based mostly on vision and force feedback, and in most cases they rely media (vision) mainly to create a frame of reference for location without the necessary tactile feedback, and visual feedback is not universally reliable. Because there is limited tactile or force (feedback) sensing, the gripper fails and objects are either misplaced or damaged. Kumar et al. [3] and Tirian [4] indicated there was potential value in real-time feedback from gripper/object handling, but both systems did not implement force monitoring systems or improved gripping surfaces, so fail-safe are not possible. Bhatia et al. highlighted gripper design highlighted the need for tactile interaction, but there were still limitations around material adaptability and force-motion transparency to give the operator device useability across variations in object surfaces. Han et al. (or similar) locate the lack of consideration for the interaction between the gripper and the object relative to navigation and placement challenges, and this added to their inefficient use of grippers.

The main research problem of this work is:

How can a warehouse robotic arm accomplish reliable gripping, adaptively sense forces, and place a wide variety of objects while maintaining precision in dynamic warehouse environments?

To address to this question, the proposed research includes the use of rubber pads to provide friction against the elements of the gripper and avoid slippage [6], FSR sensors to maintain compliant grip force to limit damage to the elements being handled [1][3], and TOF sensors to ensure the robot is placing an object correctly [2][4][7]. Using the technologies described above will allow the robotic arm to handle a variety of objects in warehouses without damage or error, improving the robustness of the overall system of a warehouse robot and ultimately increasing warehouse productivity. This research attempts to bridge the gaps identified in existing literature and create a more robust and cohesive method for pick-and-place automation.

# 2. Objectives

## 2.1. Main objectives

The general objective of this project is to create a smarter warehousing automation solution that combines cloud-based inventory management, robotic put away and picking, autonomous navigation, and QR-code based product identification to augment operational efficiency, accuracy, and scalability of supply chain management.

## 2.2. Specific Objectives

**Automate Pick-and-Place Operations** - To enable programming of the robotic arm to do stock-in by picking items from the trolley and putting them into storage racks, and stock-out by removing items from racks and putting them onto the trolley.

This eliminates repetition and saves time in operations, which reduces human intervention and errors.

**Enable Precise Releasing and Gripping Mechanisms** - To enable programming of the gripper of the robotic arm to hold items firmly while doing pick-and-place operations.

Proper handover reduces object damage or drop, and correct release ensures proper placement on racks or trolleys.

**Synchronize with Mobile Trolley Movement** - To implement a signaling mechanism between the robotic arm and the trolley so that after completion of the stock-in or stock-out, the arm signals the trolley to move to the next location.

This prevents workflow interruptions and offers free-flowing system-level coordination.

**More Efficient and Reduced Operating Costs** - To demonstrate that a robotic arm can perform warehouse activities effectively, at a reduced cost compared to industrial-grade robotic arms. Warehousing automation is made more accessible to small and medium enterprises.

## 2.3. Anticipated outcomes of objectives

The anticipated outcome of the project revolves around delivering a solid and efficient warehouse inventory automation solution.

One of its main objectives is to deliver end-to-end stock-in and stock-out automation capability to store and retrieve items without the intervention of operators, reducing delay and errors in the process. The system has high accuracy of object placement and retrieval, with no errors and maintains inventory in order and traceable.

With the use of the robotic arm with a cloud-based inventory management system, all movement of stock is updated in real time, providing warehouse managers correct and synchronized information of item location and availability.

Additionally, the integration of mobile trolley with the robotic arm is optimized to achieve maximum overall system performance and optimal efficiency to ensure smooth product transportation and efficient workflow in the warehouse.

The other expected outcome will be the existence of a cheap substitute for high-end industrial robotic arms of similar capability and precision at much lower cost, thus making top-notch warehouse automation accessible to small and medium-sized enterprises.

# 3. Methodology

## 3.1. System Diagram

Here is the conceptual system diagram of the robotic arms operations inside the smart robotic warehouse inventory system.

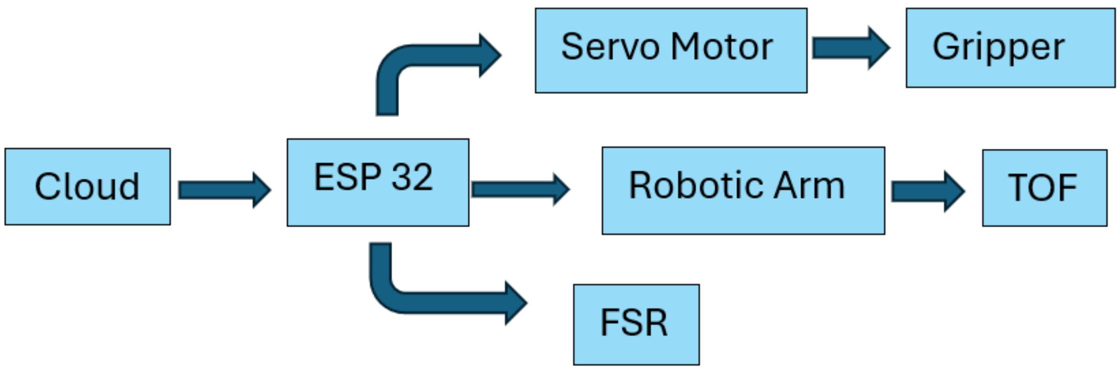


*Figure 3. 1. 1 System Diagram*

<u>Explanation</u>

The trolley is mobile within the smart warehouse system and is used for transporting goods as well as docking for stopping offloading goods. The robotic arm performs stock-in by taking products from the trolley and performing stock-out by placing products on the trolley from the racks. All operations occur in the cloud-based system (Firebase) and are processed in parallel. Thus, stock levels are always up to date in real time. All three factors together lead to an automated process which involves real time data and physical actions.

## 3.2. Steps to Take When Implementing the Project

The project will be implemented in five steps. The first one is system installation, which involves robotic arm installation and calibration as well as the controller installation. Following that is the stock-in programming stage, where the arm will be programmed to pick products from the trolley, place them in racks, and notify the trolley. Following that is the stock-out programming stage, which will program the picking operation, where products are fetched from racks, put onto a trolley, and notify the trolley. After programming is done, coordination of the arm will be put through integration testing to validate flawless control. Last, all focus will be shifting to the optimization stage, where the final changes to the robotic arm will be made.

## 3.3. Tasks and Sub-Tasks

### 3.3.1. Storage process

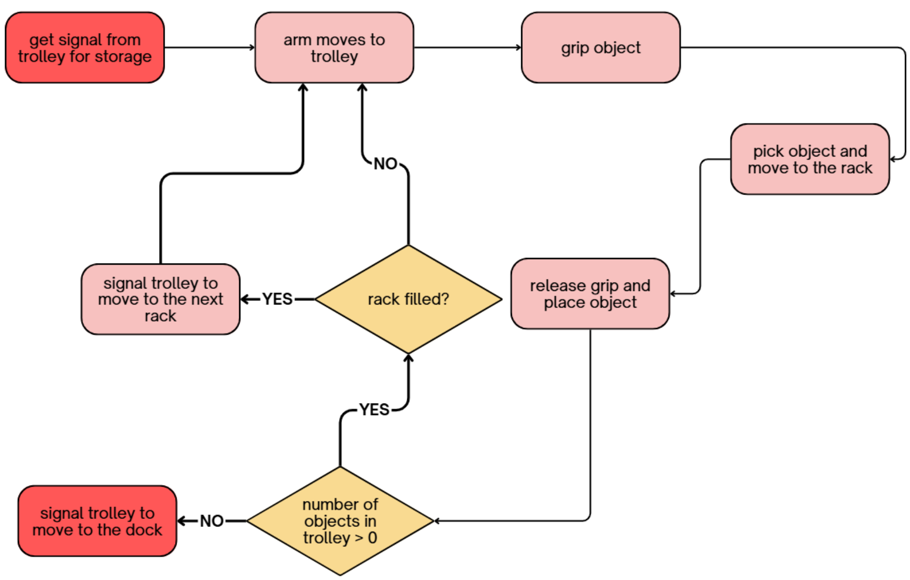


*Figure 3. 3. 1 Workflow diagram - storage process*

### 3.3.2. Retrieval process

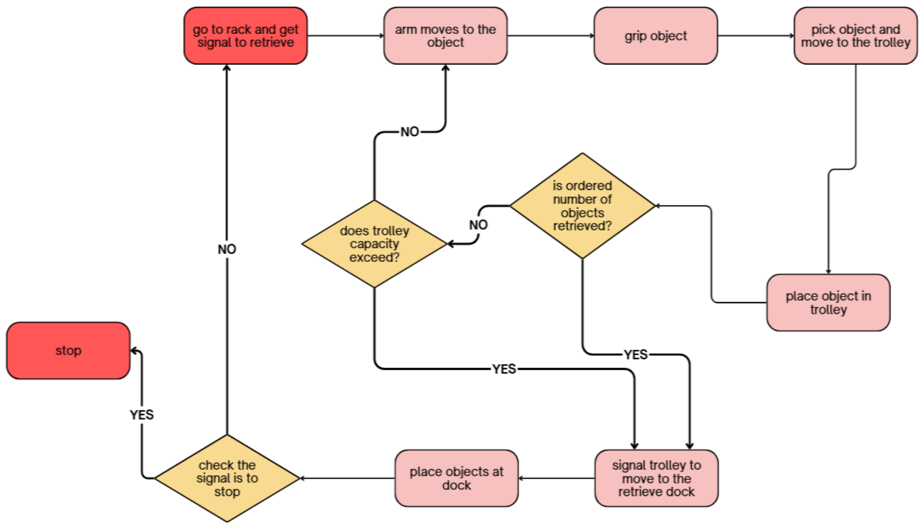


*Figure 3. 3. 2 Workflow diagram - retrieval process*

**Task 1: System Setup**

The first task is to configure the robotic arm, and to get all hardware to a stable operational state. The pre-assembled robotic arm was provided to be mounted securely at the trolley docking station accounting for the mechanical stability of the robotic arm. The robotic arm was established as operator through the compatible controller - and ESP32, Arduino, or PC - in a communication method through the use of Serial, Wi-Fi, or MQTT to determine the command.  Calibration of the components typically the gripper included confirming the range of both open and closed, grip pressure in order to pick up objects of differing weights without dropping or breaking and determining safe release points for both stock-in and stock-out.

**Task 2: Programming**

In this phase we program the robotic arm for stock-in and stock-out. For stock-in, we will program the arm to move from the trolley to pick an object and place it in the rack in a series of steps: move to object, grip, lift, move, place, and release. We will similarly program the stock-out routines to pick items from the racks and place them onto the trolley in the same sequence of steps. Error handling logic will be implemented to re-attempt gripping or releasing failure, and notifications will be delivered to the cloud system when the maximum retry attempts have been met. A notification system will also be implemented to notify the trolley when it has completed a task and includes pause and resumes for orderly coordination.

**Task 3: Testing**

The testing phase verifies and validates the system's functionality with a variety of possible cases. Unit testing verifies each routine, pick, grip, place, and release, separately, and the barcode scanning and cloud updating functions can be tested separately as well. Multi-cycle testing performs stock-in and stock-out multiple times continuously while collecting errors, times, and gripper performance. Error handling of mis-grip or failure to release can be verified by simulating these situations and observing if the system followed the proper retry and alerting. Integration testing verifies proper communication between the robotic arm, trolley, and cloud data system. In addition, proper signaling, rack assignments, and inventory updates to the associated lane's other stock products are confirmed.

**Task 4: Documentation**

A record of documentation is kept throughout the project. Technical reports describe the methodology, system design, workflows, coding logic, API calls and handling errors. Documented diagrams and charts including system architecture, process flows and Gantt charts are also prepared to show workflows and project timelines.

**Task 5: Performance Analysis**

In the last component of the project, performance analysis determines the time to execute robotic arm tasks relative to manual handling by humans. The accuracy of operations and successful rates of cloud synchronization are also reviewed to report on the overall operational efficiency and reliability of system.

# 4. Project Requirements

## 4.1. Functional Requirements

Functional requirements outline the specific goals the robotic arm system is required to accomplish. In the case of the warehouse, the following functions are requirements:

**Pick-and-Place Automation** - The robotic arm must execute stock-in (pick from trolley and place in racks) and stock-out (pick from racks and place on trolley) functions. The robotic arm should efficiently execute repetitive operations without the need for constant supervision.

**Gripper Operation** - The gripper must open and close with adequate accuracy sufficient to securely grasp and manipulate objects of varying sizes, shapes and weights. It must release the object without overly forceful motion in order to prevent damage.

**Trolley Coordination** - The robotic arm must alert the trolley after the pick-and-place cycle has been finished. The trolley should only be able to move after confirmation has been given.

**Error Handling and Recovery** - The robotic arm must take corrective action in case of improper gripping of the object.

## 4.2. Non-Functional Requirements

The robotic arm system has been developed with quality attributes ensuring competent, effective, and safe operation of warehouse activities:

**Reliability** - The system is expected to achieve a more than 95% success rate in operation, thus ensuring that errors are infrequent in pick-and-place, stock-in, and stock-out operation. Robust reliability means steady operation, even during continuous multi-cycle operations.

**Performance** - Each implementation of the operational cycles of the robotic arm (gripping, lifting, navigating, placing) is expected to be accomplished in no longer than 20 seconds, to ensure consistent productivity and flow of business in a warehouse.

**Scalability** - The design allows for the installation of robotic arms and trolleys without modifying any of the working systems. This ensures that, as more inventory comes into the warehouse or operational requirements dictate more rounds of work, the robotic arm and trolley automation set up can grow.

**Usability** - User interfaces and control mechanisms are intuitive so that warehouse operators can use the system with little, if any, training. Clear direction, visual indicators, and simple workflows make for an overall more usable operational experience.

**Security** - All communications between the robot arm, controllers, and cloud, use authenticated and encrypted communications, protecting sensitive data while preventing unauthorized access or harmful activity while ensuring satisfactory operation of the automated warehouse system.

**Maintainability** - The system has modular documentation for training and operations, which enables easy updating of operational routines and troubleshooting, along with easy adjustments of the robotic arm operations, less downtimes and improved maintainability.

## 4.3. User Requirements

**Reliably handle stock** - The robotic arm and corresponding system must repeatedly carry out stock in and stock out operations without error, appropriately picking, moving, and placing items without errors causing stock discrepancies or taking excessive amounts of time.

**Minimized manual handling** - The aim of the robotic arm is to minimize all manual handling. Operators and supervisors should only be involved in managing the operation and the robotic arm, not handling or moving inventory items. It affects workers safety, efficiency, and the overall flow of work and handling of inventory items.

**Clear/simple notifications** - When a pick task failed or a place task failed, operators should get a clear, simple notification from the robotic arm to allow them to quickly rectify any error and supervise if the automated system retries automatically or not and only if tasks have minor impact on workflow.

**Confirmation of task completion** - After any task is completed, pick, lift, or shunt, the robotic arm must signal the trolley or cloud system, confirming task completion, coordinating with other processes, and to allow the robotic arm to move the trolley, whether, by itself, or to retrieve the next item.

**Operates more quickly/effectively** - The robotic arm must operate more quickly and effectively than manual handling. The robotic arm can better utilize an optimal algorithm for FSR + TOF sensors + gripper control for safely handling inventory and place them accurately, enhancing operational performance.

## 4.4. System Requirements

### 4.4.1 Hardware Requirements

The hardware configuration for the warehouse robotic arm system includes an off-the-shelf robotic arm with a gripper mechanism for item picking and placing. The system utilizes an ESP32 microcontroller for arm control and sensor data processing. The arms utilize TOF sensors for distance measurements and FSR sensors for grip monitoring to avoid slippage or damage to items. All of this is powered by a power supply and wired so that the system may run smoothly and reliably for a long duration.

#### 4.4.2 Software Requirements

The software requirements involve a programming environment such as Arduino IDE or Python, which can be used to write and upload control routines to the ESP32 microcontroller. That development environment will allow for the development of stock-in routines and stock-out routines, handle errors, as well as to transceive updates for the status of inventory, tasks and provide two-way communication with the cloud system in real-time.

## 4.5. Use Cases

**Use Case 1: Stock-In (Storage Operation)**

The stock-in operation uses the cloud system, robotic arm, and moveable trolley as the actors. The pre-condition for this use case is that the docking station is working, and the trolley is able to move to the dock with items on it. In this particular use case, the trolley has finished moving and arrives at the docking station. The robotic arm then takes the item off the trolley and places it for barcode verification. Once the barcode is verified, the cloud assigns the item to a rack. The robotic arm then places the item in the rack assigned by the cloud. The cloud can now record the stock-in operation. After the stock-in operation has been recorded, stock is will reside on the system and inventory is updated. The main advantage from this use case is that storage operations will not require human effort, providing a satisfactory level of accuracy and maximization of labor, physically and visibly removing strain and inefficiencies from this aspect of storage operations.

**Use Case 2: Stock-Out (Retrieval Operation)**

A stock-out operation includes a robotic arm, the cloud system, the moveable trolley, and the warehouse operator. The operation begins when the warehouse operator receives a retrieval request from the dashboard, which is the pre-condition. The cloud system takes ownership of where the item is stored by finding the rack with the item and sending the robotic arm to the appropriate rack. The robotic arm will pull the item off the rack and place it on top of the trolley. When the operation is finished, the cloud system will update the stock-out record. The post-condition is that the item is ready to be dispatched. The main benefit of this use case is the automation of retrieval operations with an increase in accuracy and a reduction in manual activity.

## 4.6. Test Cases

**Test Case 1: Stock-In Functionality**

This test case seeks to test the storage of items. The steps for this procedure include putting the item on the trolley, running the stock-in function with the robotic arm, and storing the item in the Official rack. The desired outcome would be that the item is properly stored in the designated rack, confirming the storage functionality of the system.

**Test Case 2: Stock-Out Functionality**

This test case is meant to validate the retrieval process. The steps would consist of: requesting an item from the dashboard; having the robotic arm retrieve the requested item from the rack; and then having the arm place the item on the detachable trolley. The desired outcome would be that the requested item was placed on the trolley confirming that the stock-out process correctly completed the operation.

**Test Case 3: Error Recovery**

This test case aims to evaluate the system's recovery when the grip fails. A mis-grip is simulated and is followed by the retry logic built into the control logic of the robotic arm. The expected outcome is that a retry picks the object and thus demonstrates error handling.

**Test Case 4: Trolley Coordination**

This test case will evaluate the signaling of the robotic arm with the trolley. After completing a pick and place routine, the robotic arm will signal the trolley. The expected outcome is that the trolley only moves after it acknowledges the signal. Each system operates safely and is coordinated with the robotic arm and trolley systems.

## 4.7. Project Timeline

### 4.7.1. Project Flow

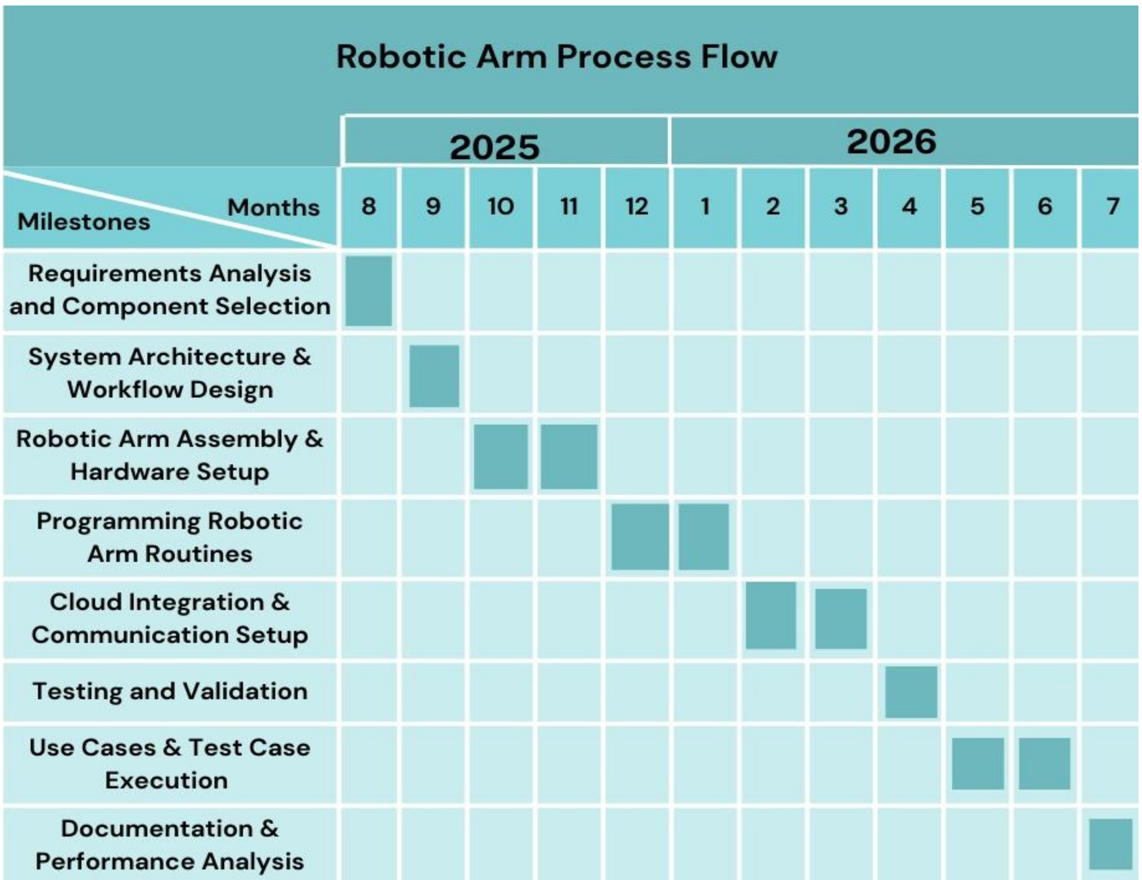


*Figure 4. 7. 1 Gantt Chart*

### 4.7.2. Communication Flow

Inputs: Tasks assigned from cloud (stock-in or stock-out).

Processing: Arm executes pick-and-place routine using pre-programmed motions.

Outputs: Object stored/retrieved, cloud database updated, trolley movement triggered.

# 5. Description of Personal & Facilities

## 5.1. Facilitators

| | |
|---|---|
| Mr. Uditha Dharmakeerthi<br>Sri Lanka Institute of Information Technology(SLIIT) | Prof. Anuradha Jayakody<br>Sri Lanka Institute of Information Technology(SLIIT) |

# 6. Description of Budget

| Item | Cost (LKR) |
| --- | --- |
| Robotic Arm kit | 10000 |
| Controller (ESP32) | 1700 |
| Gripper Mechanism | 1500 |
| Miscellaneous (power supply, wiring) | 2500 |
| Other expenses | 3000 |
| TOTAL | 18700 |

# 7. Commercialization

The smart robotic warehouse inventory management system, which expands functionality for stock-in and stock-out purposes through a robotic arm, has significant commercialization potential for logistics, retail, manufacturing, and e-commerce industries. Modern warehouses are being pushed to deliver at more speed, accuracy, and labor efficiency, but small and, to some extent, medium-sized enterprises will not be able to afford high-cost, fully automated systems currently on the market. By incorporating cloud-based inventory management with reliance on a ready-made robotic arm on mobile trolleys, this project has provided a low-cost but capable automation solution and is almost perfectly scalable with the warehouse operational capacity.

The robotic arm subsystem can be commercialized as a module within the full warehouse automation system along with trolleys, barcode scanners, and dashboards.

Its commercialization strategy can start with pilot deployments for new small-scale distribution centers and then look to expand towards mid-level warehouses that are implementing modular, cost-effective automation.

## 7.1. Target Audience

Small to Medium-Sized Warehouses – companies that want to automate their warehousing operations but cannot purchase expensive industrial arms.

E-commerce Startups – companies that use small fulfillment centers and want fast, accurate, and affordable handling of their inventory.

Retail Chains – companies with back-end storage facilities where automation better organizes stock control processes.

Educational Institutions and Research Labs – uses might include teaching, demonstration of technical features, and experimenting with robotics and automation.

## 7.2. Competitive Advantage

Cost – Utilizes a common ready-built robotic arm and ESP32-based control for a large cost reduction as compared to industrial robotic arms.

Modular – The arm can be used in a stand-alone mode or in combination with other components in a smart warehouse system.

Easy Integration – Because it works through Wi-Fi and is platform open source, it allows easy connection to cloud/databases.

Versatility – It can support stock-in and stock-out tasks, unlike some low-cost arm solutions are only demonstration lessons.

Scalability – Multiple arms can be deployed to different parts of a warehouse to complete parallel operations.

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